MILITARY SPECIFICATION SHEET

ELECTRON TUBES, RECEIVING

TYPES 6SH7 AND 12SH7 1/ 2/
a b

Ⓡ

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pentode, RF sharp cutoff

Outline a, b --- 8-1 (EIA) Ⓡ

Base a, b --- B8-21

Envelope a, b --- MT8

Cathode a, b --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8
Element	a, b ---	sh	h	k, g3	g1	k	g2	h	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:		Ef	Eb	Ec1	Ec2	Esh	Pg2	Pp	Ehk	Alt
Unit:		V	Vdc	Vdc	Vdc	Vdc	W	W	v	ft
Maximum:	a	6.9	330	---	165	---	0.7	3.3	100	10,000
Minimum:	a	5.7	---	---	---	---	---	---	---	---
Maximum:	b	13.8	330	---	165	---	0.7	3.3	100	10,000
Minimum:	b	11.3	---	---	---	---	---	---	---	---
TEST CONDITIONS:	a	6.3	250	-1	150	0	---	---	---	---
	b	12.6	250	-1	150	0	---	---	---	---

GENERAL:

Qualification - Required

1/ To identify those tests that are applicable to a given type or to several types; tube types are designated by letters.
2/ See note 4.

Ⓡ denotes changes

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:30Z
Check the source to verify that this is the currentversion before use.

METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1231	Emission	a, b	Eb = Ec1 = Ec2 = 20 Vdc (see note 2)	0.65	II	Is	95	---	mAdc
1266	Total grid current	a, b	Ec1 = -1.5 Vdc (see note 2)	0.65	II	Ic1	0	-1.0	µAdc
1256	Electrode current (anode)	a, b		0.65	II	Ib	8.2	13.4	mAdc
1256	Electrode current (cathode)	a, b	Eb = Ec2 = 150 Vdc; Ec1 = -6.5 Vdc (see note 1)	0.65	II	Ik	0	50	µAdc
1306	Transconductance	a, b		0.65	II	Sm	3,920	5,880	µmhos
1201	Short and discontinuity detection	a, b		0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	a, b	Rp = 2,000 ohms	6.5	S3	Ep	---	400	mVac
1301	Heater current	a		6.5	S3	If	275	325	mA
		b		6.5	S3	If	138	162	mA
Ⓑ 1336	Heater-cathode leakage	a, b		6.5	S3	Ihk	---	20	µAdc
1256	Electrode current (screen)	a, b		6.5	S3	Ic2	2.0	6.0	mAdc
1331	Direct-interelectrode capacitance	a, b		6.5	S3	Cgp	---	0.0035	pF
						Cin	6.8	10.2	pF
						Cout	4.9	9.1	pF
1101	Secureness of base and base insert	a, b		6.5	S3	---	---	---	---
1111	Base pin solder depth	a, b	(See note 3)	6.5	S3	---	---	---	---
1211	Insulation of electrodes	a, b		4.0	S3	R	10	---	Meg
1105	Permanence of marking	a, b		---	---	---	---	---	---
	Quality conformance inspection, part 3								
1501	Intermittent life test provisions (500 hours)	a, b	Group A; Ehk = 100 V	---	---	---	---	---	---
---	Life-test end point (intermittent)(500 hours)	a, b	Transconductance	---	---	Sm	3,180	---	µmhos

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:30Z
Check the source to verify that this is the current version before use.

NOTES:

1. Tie grid No. 2 to anode at socket.
2. This test to be performed at the conclusion of the holding period.
3. This test applies to flexible leads as well as to rigid leads.
4. Tube type 12SH7GT is hereby deleted from this TSS. Tubes for replacement purposes will be procured to EIA data, Release Number 282, Type 12SH7GT. (R)

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

(Project 5960-2211-20)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:30Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:30Z
Check the source to verify that this is the current version before use.